

Implement a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, emission-responsive charging, and the rebound effect.

Executive Technology Brief

Domain: ml_data

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'fuel inefficiency' concerning car. User-submitted engineering problem (session input): "most fuel efficient car on earth". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

Mechanism: Dynamic pricing and optimization of fuel consumption based on real-time data analytics and machine learning algorithms.

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: Implement a smart fuel pricing system that adjusts fuel prices in real-time based on factors such as road congestion, emission-responsive charging, and the rebound effect.

Governing Model: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: APPLICABLE; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: $P(x|y)$.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: APPLICABLE; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: $n, f(x_i)$.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: CONDITIONAL; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.